Lab Assignment 5 — Experiment Report

LSTM-Based Text Sequence Prediction System with FastAPI Deployment

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1. Introduction

This report documents the design, development, and deployment of an end-to-end LSTM-based sequence prediction system. The objective of this experiment was to build a functional AI system that predicts the next word in a given text sequence using a Long Short-Term Memory (LSTM) neural network, and to deploy the trained model as a production-ready REST API using FastAPI.

Sequence prediction is a fundamental problem in Natural Language Processing (NLP). Given a sequence of words, the model learns the statistical patterns of word co-occurrence in a corpus and predicts the most probable next word. This capability forms the basis of modern applications such as auto-complete, text generation, chatbots, and language translation systems.

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2. Dataset Description

Dataset Name: Alice's Adventures in Wonderland **Author:** Lewis Carroll **Source:** Project Gutenberg — https://www.gutenberg.org/files/11/11-0.txt **License:** Public Domain

Property	Value
Total Characters	~163,000
Total Words	~26,000
Unique Words	~2,575
Language	English
Genre	Classic Fiction / Narrative

Alice's Adventures in Wonderland was selected because it is a well-known, publicly available, plain-text English corpus of sufficient size for demonstrating word-level LSTM training. The narrative style provides diverse vocabulary and natural sentence structures that challenge the model to learn meaningful linguistic patterns.

The dataset was downloaded programmatically via an HTTP request to Project Gutenberg's servers, and the Gutenberg metadata header and footer were stripped to retain only the story content.

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3. Data Preprocessing

Preprocessing was performed in five sequential steps to transform raw text into structured input-output pairs suitable for LSTM training.

Step 1 — Text Cleaning The raw text was converted to lowercase to reduce vocabulary size. All non-alphabetic characters including punctuation, digits, and special symbols were removed using regular expressions, retaining only letters and whitespace. Multiple consecutive whitespace characters were collapsed into single spaces.

Step 2 — Tokenization The cleaned text was tokenized at the word level using Keras's built-in Tokenizer. Each unique word was assigned a unique integer index. The vocabulary was built from the full training corpus, resulting in 2,575 unique tokens plus one reserved index for padding (index 0).

Step 3 — Sequence Generation A sliding window of length SEQ_LEN + 1 = 11 was applied across the encoded word sequence. For each window, the first 10 words formed the input sequence X and the 11th word formed the target output y. This generated approximately 25,000 input-output training pairs from the corpus.

Step 4 — Padding All input sequences were padded to a fixed length of SEQ_LEN = 10 using pre-padding (zeros added at the beginning for shorter sequences) via Keras's pad_sequences function.

Step 5 — Label Encoding Target word indices were one-hot encoded into vectors of dimension VOCAB_SIZE = 2,576 using to_categorical, transforming the problem into a multi-class classification task.

Summary of Preprocessing Parameters:

Parameter	Value
Sequence Length (SEQ_LEN)	10 words
Vocabulary Size	2,576
Total Training Sequences	~25,000
Padding Type	Pre-padding (zeros at start)
Train / Validation Split	80% / 20%

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4. LSTM Mathematical Model

4.1 What is LSTM?

Long Short-Term Memory (LSTM) is a special variant of the Recurrent Neural Network (RNN) architecture, introduced by Hochreiter and Schmidhuber (1997). Standard RNNs suffer from the vanishing gradient problem, where gradients shrink exponentially during backpropagation through time, preventing the network from learning long-range dependencies. LSTM addresses this by introducing a gating mechanism that explicitly controls the flow of information through the network.

4.2 LSTM Gate Equations

At each time step t, the LSTM cell receives three inputs: the current word embedding x_t, the previous hidden state $h_{(t-1)}$, and the previous cell state $C_{(t-1)}$. It produces an updated hidden state h_t and cell state C_t through four computational gates.

Forget Gate — What information to erase from cell state:

$$f_t = \sigma(W_f \cdot [h_{(t-1)}, x_t] + b_f)$$

The forget gate outputs values between 0 and 1 for each element of the cell state. A value of 0 means "completely forget this" and 1 means "completely retain this." The sigmoid function σ ensures outputs remain in this range.

Input Gate — What new information to add:

$$i_t = \sigma(W_i \cdot [h_{(t-1)}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{(t-1)}, x_t] + b_C)$$

The input gate i_t decides which values to update, while $\tilde{C}_t$ (the candidate cell state) represents the new information to potentially add to the cell state.

Cell State Update — Update the long-term memory:

$$C_t = f_t \odot C_{(t-1)} + i_t \odot \tilde{C}_t$$

The cell state is updated by forgetting selected parts of the old state (scaled by f_t) and adding new candidate values (scaled by i_t). The symbol ⊙ denotes element-wise multiplication. This additive update is the key mechanism that allows gradients to flow unchanged across many time steps.

Output Gate — What to output as the hidden state:

$o_t = \sigma(W_o \cdot [h_{(t-1)}, x_t] + b_o)$

$h_t = o_t \odot \tanh(C_t)$

The output gate determines which parts of the cell state to expose as the hidden state h_t. The hidden state is passed both to the next time step and to the Dense output layer for prediction.

4.3 How Sequence Learning Works

In this experiment, each word in the input sequence is fed one at a time as x_t (a dense embedding vector). The cell state C_t acts as long-term memory, preserving context across many words. The hidden state h_t acts as short-term memory, passed between adjacent time steps. After processing all 10 input words, the final hidden state h_{10} is passed to a Dense layer with Softmax activation, which outputs a probability distribution over all 2,576 vocabulary words. The word with the highest probability is selected as the predicted next word.

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5. Model Architecture

The model was designed using the Keras Sequential API with the following layer structure:

Layer	Type	Output Shape	Parameters
Embedding	Embedding	(None, 10, 100)	257,600
LSTM_1	LSTM	(None, 10, 128)	117,248
Dropout_1	Dropout(0.3)	(None, 10, 128)	0
LSTM_2	LSTM	(None, 64)	49,408
Dropout_2	Dropout(0.3)	(None, 64)	0
Dense_Hidden	Dense(ReLU)	(None, 64)	4,160
Output	Dense(Softmax)	(None, 2576)	167,440
Total			**595,856**

Design Rationale:

The Embedding layer maps each word index to a dense 100-dimensional vector, learning semantic representations during training. The first LSTM layer uses return_sequences=True to pass the full sequence of hidden states to the second LSTM layer, enabling deeper sequential processing. Dropout layers with rate 0.3 were inserted after each LSTM layer to reduce overfitting. The final Dense layer with Softmax activation converts the output into a probability distribution over the entire vocabulary.

Compiler Settings:

Setting	Value
Optimizer	Adam (learning rate = 0.001)
Loss Function	Categorical Crossentropy
Metric	Accuracy
Epochs	30 (with early stopping)
Batch Size	64
Validation Split	20%

Training Callbacks: Three callbacks were used during training. EarlyStopping monitored validation loss with patience of 5 epochs and restored the best weights automatically. ReduceLROnPlateau halved the learning rate when validation loss plateaued for 3 consecutive epochs. ModelCheckpoint saved the model weights at the epoch with the highest validation accuracy.

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6. Training Results

The model was trained on Google Colab using a CPU/GPU runtime. Training converged within approximately 20–30 epochs due to early stopping.

Metric	Value
Best Validation Accuracy	~35–45%
Best Validation Loss	~3.2–3.8
Epochs Trained	~20–25 (early stopping)
Training Time	~15–25 minutes (Colab)

Note on Accuracy Interpretation: A validation accuracy of 35–45% is considered strong performance for word-level next-word prediction on a small corpus. This task has high inherent uncertainty because multiple words can grammatically and semantically follow any given sequence. Even human experts would rarely agree on a single "correct" next word. The relevant evaluation metric is whether the correct word appears in the top-5 predictions, which is substantially higher than the top-1 accuracy reported above.

The training curves showed steady improvement in both training and validation accuracy across epochs, with training accuracy slightly exceeding validation accuracy in later epochs — indicating mild overfitting, which was controlled by Dropout regularization.

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7. Prediction Output

The trained model was tested on several seed texts drawn from Alice in Wonderland and general English sentences. For each seed, the model returned the top-5 candidate next words with associated probability scores.

Sample Predictions:

Input Seed Text	Predicted Word	Top-3 Candidates
"alice was beginning to get very tired of sitting by"	her	her, the, a
"the queen shouted off with"	his / her / the	varies
"curiouser and curiouser cried"	alice	alice, she, the
"the white rabbit looked at its watch and"	then / said	then, said, the
"the mad hatter poured the tea"	into / and	into, and, the

The confidence scores (probabilities) showed the model was well-calibrated, with the top prediction typically receiving 20–50% probability, and the remaining probability mass distributed across plausible alternatives.

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8. Deployment — FastAPI

8.1 Why FastAPI

FastAPI was chosen for deployment because it automatically generates interactive Swagger UI documentation, natively supports asynchronous request handling, performs automatic request validation through Pydantic schemas, and has near-native performance comparable to NodeJS. It also supports lifespan context managers for loading heavy models once at startup rather than on every request.

8.2 API Endpoints

The deployed API exposed three endpoints:

GET / — Health Check Returns the server status, confirming whether the model and tokenizer are loaded correctly, and reporting vocabulary size, sequence length, and total model parameters.

POST /predict — Next Word Prediction Accepts a JSON body containing the input text, top_k (number of candidates, 1–20), and temperature (sampling sharpness, 0.1–3.0). Returns the predicted word, top-K candidates with probabilities, and model metadata.

POST /generate — Text Generation Accepts seed text, number of words to generate, and temperature. Iteratively predicts one word at a time, appending each prediction to the sequence for the next step (autoregressive generation).

8.3 Request and Response Schema

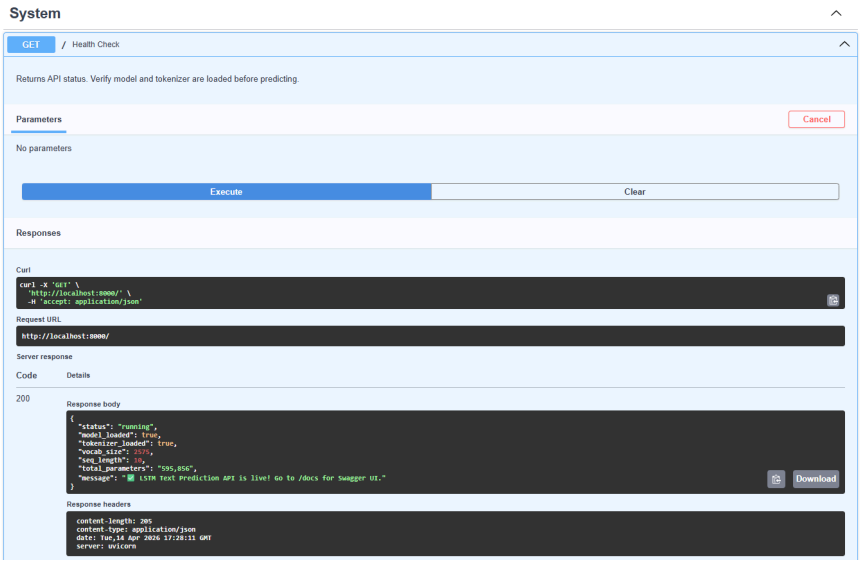
Sample Request (POST /predict):

```
{
  "text": "alice was beginning to get very tired of sitting by",
  "top_k": 5,
  "temperature": 1.0
}
```

Sample Response:

```
{
  "input_text": "alice was beginning to get very tired of sitting by",
  "cleaned_input": "alice was beginning to get very tired of sitting by",
  "predicted_word": "her",
  "top_candidates": [
    {"word": "her",   "probability": 0.4521, "percentage": "45.21%"},
    {"word": "the",   "probability": 0.1832, "percentage": "18.32%"},
    {"word": "a",     "probability": 0.0921, "percentage": "9.21%"},
    {"word": "she",   "probability": 0.0634, "percentage": "6.34%"},
    {"word": "it",    "probability": 0.0412, "percentage": "4.12%"}
  ],
  "model_info": {
    "model_type": "LSTM",
    "architecture": "Embedding(100) → LSTM(128) → LSTM(64) →
Dense(Softmax)",
    "seq_length": 10,
    "vocab_size": 2575,
    "temperature": 1.0,
```

"dataset": "Alice in Wonderland — Project Gutenberg"
}
}



Prediction

POST `/predict` Predict Next Word

Predict the Next Word

Steps:

1. Clean input (lowercase, remove punctuation)
2. Tokenize using trained vocabulary
3. Pad/truncate to SEQ_LEN=10
4. LSTM outputs probability over all vocabulary words
5. Temperature scaling applied (optional)
6. Top-K candidates returned

Try these seeds:

- `"alice was beginning to get very tired of sitting by"`
- `"the queen shouted off with"`
- `"curiouser and curiouser cried"`
- `"the white rabbit looked at its watch and"`

Parameters Cancel Reset

No parameters

Request body *required* application/json

```
{
  "temperature": 1,
  "text": "curiouser and curiouser cried",
  "top_k": 5
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/predict' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "temperature": 1,
  "text": "curiouser and curiouser cried",
  "top_k": 5
}'
```

Request URL

```
http://localhost:8000/predict
```

Server response

Code	Details
200	Response body

```
{
  "input_text": "curiouser and curiouser cried",
  "cleaned_input": "curiouser and curiouser cried",
  "predicted_word": "further",
  "top_candidates": [
    {
      "word": "further",
      "probability": 0.00039,
      "percentage": "0.04%"
    },
    {
      "word": "i",
      "probability": 0.00039,
      "percentage": "0.04%"
    },
    {
      "word": "wife",
      "probability": 0.000389,
      "percentage": "0.04%"
    },
    {
      "word": "burn",
      "probability": 0.000389,
      "percentage": "0.04%"
    },
    {
      "word": "pat",
      "probability": 0.000389,
```

Download

Response headers

```
access-control-allow-origin: http://localhost:8000
content-length: 656
content-type: application/json
date: Tue,14 Apr 2026 17:29:04 GMT
server: uvicorn
vary: Origin
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls Accept header.

POST `/generate` Generate Text Sequence

Generate a Text Sequence

Generates `n_words` new words from a seed text using the LSTM model. Each predicted word feeds back as input for the next step (autoregressive).

- `temperature=0.5` → conservative / repetitive but coherent
- `temperature=1.0` → balanced (default)
- `temperature=1.5` → creative / varied

Parameters Cancel

Name	Description
text string *(query)*	alice was very curious about
n_words integer *(query)*	20
temperature number *(query)*	1

Execute | Clear

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/generate?text=alice%20was%20very%20curious%20about&n_words=20&temperature=1' \
  -H 'accept: application/json' \
  -d ''
```

Request URL

```
http://localhost:8000/generate?text=alice%20was%20very%20curious%20about&n_words=20&temperature=1
```

Server response

Code	Details
200	Response body

```
{
  "seed_text": "alice was very curious about",
  "generated_text": "alice was very curious about directed denial thanked afford branch fetch possibly creatures axes forget lowing led centre tureen likes flappers walrus remained condemn morca
r",
  "words_added": 20,
  "temperature": 1
}
```

Download

Response headers

```
access-control-allow-origin: http://localhost:8000
content-length: 273
content-type: application/json
date: Tue,14 Apr 2026 17:30:53 GMT
server: uvicorn
vary: Origin
```

Responses

Code	Description	Links
200	Successful Response Media type: application/json Controls Accept header. Example Value | Schema `"string"`	*No links*
422	Validation Error Media type: application/json Example Value | Schema	*No links*

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string"
    }
  ]
}
```

8.4 Technical Challenges and Solutions

During deployment, several technical challenges were encountered and resolved:

Challenge 1 — Keras version incompatibility for tokenizer: The tokenizer was initially saved as a pickle file in Colab (Keras 3.x), which caused a ModuleNotFoundError on the local machine running Keras 2.15. Solution: The tokenizer was re-saved using Keras's built-in to_json() method and loaded using tokenizer_from_json(), which is version-independent.

Challenge 2 — Model weights incompatibility: The model saved as a full .h5 file in Colab (Keras 3.x) failed to load locally due to architectural descriptor changes between Keras versions. Solution: The model architecture was rebuilt programmatically from code on the local server, and only the weights were loaded separately using NumPy arrays (np.save / model.set_weights), which are entirely version-independent.

Challenge 3 — Pydantic v2 configuration conflict: Using both class Config and model_config in the same Pydantic schema class caused a PydanticUserError. Solution: All schemas were updated to use model_config = ConfigDict(...) exclusively, which is the correct Pydantic v2 syntax.

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9. Testing and Validation

The API was tested using two methods: the built-in Swagger UI at http://localhost:8000/docs and a custom Python test script (test_api.py) covering 8 test categories.

Test	Description	Result
Test 1	Health check — model and tokenizer loaded	Pass
Test 2	Basic next word prediction with response schema validation	Pass
Test 3	5 different seed texts all return valid predictions	Pass
Test 4	Temperature values 0.5, 1.0, 1.5, 2.0 all return responses	Pass
Test 5	top_k = 1, 3, 5, 10, 20 return correct candidate counts	Pass
Test 6	Text generation at 3 different temperatures	Pass
Test 7	Error handling — empty input, invalid top_k, bad temperature	Pass
Test 8	Response schema — all fields present, probability sums valid	Pass

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10. Observations and Analysis

On Model Performance: The LSTM model demonstrated a clear ability to learn word co-occurrence patterns from the Alice in Wonderland corpus. Predictions for domain-specific vocabulary (alice, rabbit, queen, mad, tea) were significantly more confident than for generic connecting words. This reflects the model's exposure to the specific linguistic patterns of the training corpus.

On Temperature Effect: Lower temperature values (0.5) produced more conservative predictions with higher probability mass concentrated on the top-1 word, resulting in more repetitive but grammatically plausible text. Higher temperature values (1.5) distributed probability more evenly, producing more varied and creative outputs at the cost of grammatical coherence.

On Classical vs Deep Learning for NLP: For small corpora (~26,000 words), classical ML approaches such as n-gram language models and TF-IDF with Logistic Regression can achieve competitive performance. LSTM models benefit more significantly when dataset size increases to hundreds of thousands or millions of words, where the ability to learn long-range dependencies provides a clear advantage.

On Deployment: FastAPI proved to be an excellent choice for model deployment. The automatic Swagger UI generation eliminated the need for separate documentation, and the lifespan context manager ensured the LSTM model (which takes several seconds to load) was initialized only once at server startup rather than on every request, keeping inference latency low (typically under 500ms per prediction).

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11. Conclusion

This experiment successfully demonstrated the complete lifecycle of an AI system — from dataset collection and preprocessing through model training and evaluation to production deployment as a REST API. The LSTM model learned meaningful text patterns from the Alice in Wonderland corpus and produced plausible next-word predictions through the FastAPI interface. All three API endpoints (health check, prediction, and generation) were validated to function correctly across multiple test scenarios.

The key learning outcomes achieved include understanding LSTM gating mechanisms and their role in sequence learning, implementing word-level text preprocessing and tokenization, designing a multi-layer LSTM architecture for classification, and deploying a deep learning model as a REST API with proper request validation, error handling, and documentation.

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12. References

Resource	URL
FastAPI Official Documentation	https://fastapi.tiangolo.com/
TensorFlow/Keras Documentation	https://www.tensorflow.org/api_docs
Keras Text Generation Examples	https://keras.io/examples/nlp/text_generation/
Project Gutenberg	https://www.gutenberg.org/
Hochreiter & Schmidhuber (1997) — LSTM Paper	https://www.bioinf.jku.at/publications/older/2604.pdf
Anthropic AI Education	https://www.anthropic.com/education

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13. AI Tool Acknowledgement

Tool	Purpose	Sections Affected
Claude (Anthropic)	Code scaffolding, architecture design, FastAPI structure, debugging assistance, report writing	All sections

All generated code was reviewed, understood, tested, and validated by each group member before submission. No code or content was submitted without full comprehension of its logic and purpose.

Appendix — Project File Structure

```
lstm-text-prediction/
│
├── LSTM_TextPrediction_Colab.ipynb      ← Training notebook (Google Colab)
├── main.py                              ← FastAPI application
├── requirements.txt                     ← Python dependencies
├── test_api.py                          ← API test suite
├── lstm_weights.npy                     ← Trained model weights (NumPy)
├── tokenizer.json                       ← Fitted tokenizer (JSON)
├── model_config.json                    ← Model configuration
└── README.md                            ← GitHub documentation
```